Duc-Huy Do

duchuydo.ddh@gmail.com

Seattle, WA | +1 (541) 604-2147 | linkedin.com/in/duchuydo | Authorized to work in the US with security clearance

SUMMARY

Senior software engineer and tech lead with 10+ years building cloud infrastructure, microservices, and reliability systems, leading teams of different sizes across industries. Currently lead a team of 6 at TikTok supporting content moderation pipelines handling 2B+ videos/day, building availability, disaster recovery, and alerting systems, and cloud cost optimization saving $3M+/year. Previously led cloud migration strategy and platform engineering at Deloitte and a developer portal serving 800+ users at GE Healthcare.

EXPERIENCE

TikTok, Seattle, WA — Sep 2023 – Present

Senior Software Engineer (Tech Lead)

- Led a team of 6 engineers supporting content moderation pipelines across content types, handling 2B+ videos daily.
- Built availability, performance, and disaster recovery failover systems for production with zero fatal outages, using LLMs to audit alarms, add missing ones, tune thresholds, decouple dependencies, and remove duplicates.
- Built a capacity allocation and cost-optimization web application for 1K+ compute instances, saving $3M+ annually.
- Developed an alarm handling and management system for production applications, cutting alert noise by 35%, MTTA by 20%, and MTTR by 50%+.

  Tech: Go, Node.js, TypeScript, SQL, RDS, Redis, Eventbus, Hertz, Kitex, LLM

Deloitte, Seattle, WA — Jan 2021 – Aug 2023

Senior Software Engineer (Tech Lead)

- Led cloud suitability assessments and migration strategy for client applications on AWS, defining migration patterns adopted across 4 engagements.
- Led a team of 5 engineers building infrastructure monitoring for compute instances running client web applications, covering 120+ instances.
- Built a platform for developing and deploying containerized, cloud-native web applications across AWS and GCP, reducing deployment time by over 50%.

  Tech: AWS, GCP, Terraform, Python, Docker, Kubernetes

GE Healthcare, Seattle, WA — Dec 2017 – Oct 2020

Senior Software Engineer (Tech Lead)

- Led a team of 8 engineers building a self-service developer portal aligned with healthcare quality management processes, serving 800+ registered users, 30 partners, and 300 published catalog items.
- Designed and implemented a content management web application on an API-driven serverless architecture with containerized services, supporting over a dozen technical writer teams.
- Architected cloud infrastructure following AWS Well-Architected best practices, with templated build and release automation that reduced deployment time by 80%.

  Tech: AWS (Lambda, API Gateway, ECS/EKS, CloudFormation), Terraform, Python, Node.js, Angular, Drupal, Apigee, Docker, Kubernetes

TSRI, Kirkland, WA — Jan 2016 – Dec 2017

Software Engineer

- Built a cloud infrastructure pipeline for Java-based web applications and developed a full-stack document auditing application leveraging cloud infrastructure.
- Developed backend services for source code transformation.

  Tech: AWS, Java, Python, C++, Angular, Node.js, Tomcat, SQL, ANTLR

Earlier: Software Engineer, Tyemill (Jun – Sep 2015) — backend services for a financial portfolio management web app (Ruby, Heroku).

SKILLS

Languages: Go, Python, Java, TypeScript/JavaScript, SQL
Cloud & Infrastructure: AWS, GCP, Kubernetes, Docker, Terraform, CloudFormation, Apache HTTP Server, Tomcat
Frameworks: Node.js, Angular, Hertz, Kitex
Data: PostgreSQL, MySQL, NoSQL, Redshift, RDS, Redis
Practices: Microservices, serverless, web applications, infrastructure as code, site reliability, AI integration, Agile/Scrum, SDLC
Certificates: AWS Certified Developer Associate, CompTIA Security+

EDUCATION

University of Washington — M.S. in Computer Science
University of Washington — B.S. in Mathematics
Languages: French (native), Vietnamese (native)